

Mathematics

Trial Higher School Certificate Examination

2010

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1 – 10
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Total Marks – 120**Attempt Questions 1-10****All Questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 Marks)

Use a Separate Booklet

Marks

a) Evaluate $\frac{\sqrt{5^2+144}}{13-6}$ to two decimal places.

1

b) Evaluate $\int_1^5 (3x - 7) dx$

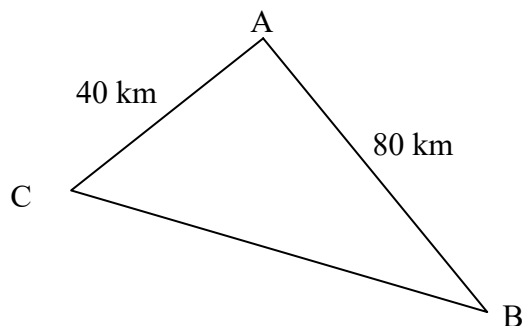
1

c) Rationalise the denominator and simplify:

2

$$\frac{1}{2-\sqrt{3}} - \frac{1}{2+\sqrt{3}}$$

d) Three towns form a triangle. Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240° .

i) Find the area enclosed by the 3 towns

2

ii) Using the cosine rule, find the distance to the nearest kilometre between Town B and Town C

2

e) Express the following as a single fraction

$$\frac{5}{2a+6} + \frac{a}{a^2-9}$$

2

f) Solve $|2x+5| < 3$

2**End of Question 1**

Question 2 (12 Marks)	Use a Separate Booklet	Marks
a) Differentiate with respect to x :		
i) $3x^2 + 7$		2
ii) $4x^2 e^{3x^3}$		2
iii) $\frac{\pi \cos x}{x^2}$		2
b) Find $\int \frac{dx}{3x+5}$		2
c) On a diagram, indicate the region where the following inequalities hold simultaneously: $y + 1 \geq 0, \quad x + y - 2 \leq 0 \quad \text{and} \quad x \geq 2$		2
d) Find the obtuse angle in degrees and minutes, that a line with gradient -2.5 makes with the positive x axis.		2

End of Question 2

Question 3 (12 Marks)	Use a Separate Booklet	Marks
a) Find $\lim_{x \rightarrow -3} \frac{x^2 + 8x + 15}{x + 3}$		2
b) Evaluate $\int_0^{\frac{\pi}{6}} (x^2 + \sin 2x) dx$. Leave your answer in exact form.		2
c) i) On the same set of axes, sketch the functions $y = 4x - x^2$ and $y = 3$		2
ii) Find the area contained between these two curves		2
d) Determine if the line $x + y + 3 = 0$ is a tangent to the parabola $y = 2x^2 + 3x - 1$		2
e) For the curve $y = \sin \pi x$, state the period and amplitude		2

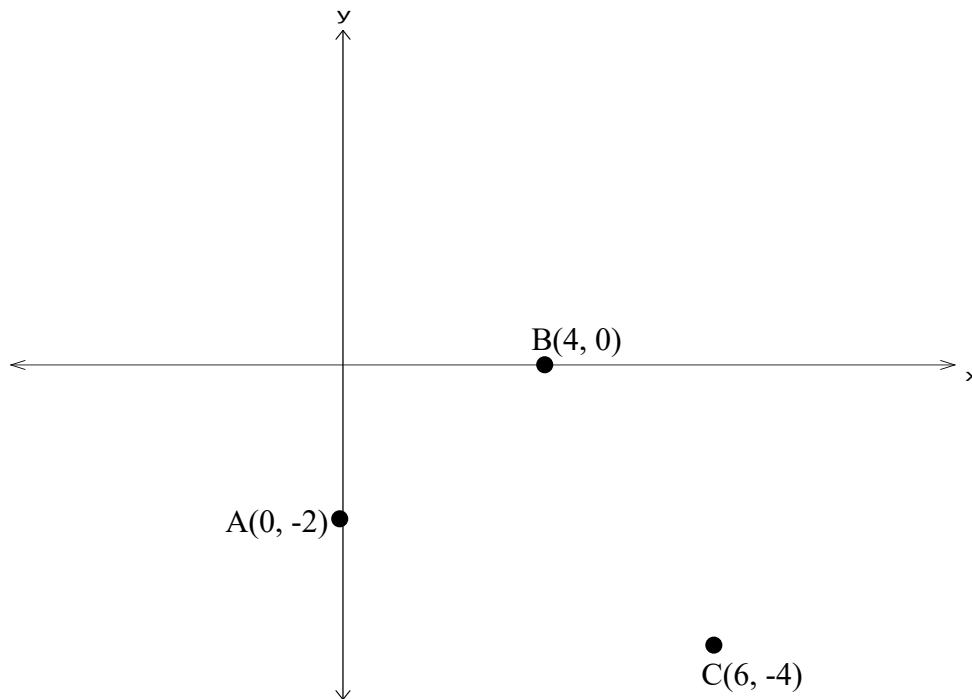
End of Question 3

Question 4 (12 Marks)

Use a Separate Booklet

Marks

- a) The coordinates of the points A , B and C are $(0, -2)$, $(4, 0)$ and $(6, -4)$ respectively.



- | | | |
|--------------------------------|--|---|
| (i) | Find the length AB , and the gradient of AB . | 2 |
| (ii) | Show that the equation of the line L , drawn through C parallel to AB , is $x - 2y - 14 = 0$. | 1 |
| (iii) | Find the coordinates of D , the point where L intersects the x -axis. | 1 |
| (iv) | Find the perpendicular distance of the point B from the line L . | 1 |
| (v) | Find the area of the quadrilateral $ABDC$. | 2 |
| b) For the arithmetic sequence | | |
| 2, 7, 12, 17, | | |
| i) | Find the general term T_n | 1 |
| ii) | Find the 23 rd term | 1 |
| iii) | Find the sum of the first 47 terms | 1 |
| c) | Find the exact value of x such that $\sec x + 1 = 3$ where $0 \leq x \leq \frac{\pi}{2}$ | 2 |

End of Question 4

Question 5 (12 Marks)	Use a Separate Booklet	Marks
a) Calculate the area of the region enclosed by the graph of $y = \cos 2x$ the x axis and the lines $x = 0$ and $x = \frac{\pi}{4}$		2
b) The roots of the equation $2x^2 - 7x + 12 = 0$ are α and β Find:		
i) $\alpha + \beta$		1
ii) $\alpha\beta$		1
iii) $\frac{1}{\alpha} + \frac{1}{\beta}$		2
iv) $\alpha^2 + \beta^2$		2
c) A pendulum consisting of a bob and a long string attached to a fixed point is set swinging with an initial arc of 40cm. If each subsequent oscillation is $\frac{5}{6}$ of the preceding one, Find the total distance travelled by the bob before it comes to rest.		2
d) The gradient function of a curve is $y' = \frac{4x}{x^2 + 1}$ and the curve passes through the point $(0, e)$. Find the equation of the curve.		2

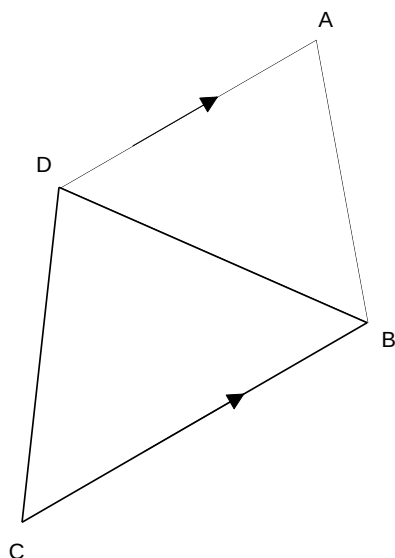
End of Question 5

Question 6 (12 Marks)

Use a Separate Booklet

Marks

a)

**3**

Given $AD = AB$, $DB = DC$, $AD \parallel BC$ and $\angle DAB = 100^\circ$.
Copy or trace the diagram into your answer booklet.

Find $\angle BDC$ giving reasons for each step.

b) Let $f(x) = x^3 - 6x^2$

- | | | |
|------|--|----------|
| i) | Find the coordinates where the curve crosses the axes. | 2 |
| ii) | Find the coordinates of any stationary points and determine their nature. | 2 |
| iii) | Find the coordinates of any points of inflexion. | 1 |
| iv) | Sketch the curve $y = f(x)$, indicating clearly the intercepts and any stationary points and points of inflexion. | 3 |
| v) | For what values of x is $y = f(x)$ increasing. | 1 |

End of Question 6

Question 7 (12 Marks)

Use a Separate Booklet

Marks

- a) i) Given $f(x) = \sqrt{4 - x^2}$, copy and complete the table of values to 3 decimal places.

1

x	0	0.5	1	1.5	2
$f(x)$					

- ii) Hence evaluate an approximation for $\int_0^2 \sqrt{4 - x^2} dx$ using Simpson's rule with 5 function values.

2

- b) A pendulum on a grandfather clock is 50 cm long. When it swings the maximum length of the arc it makes is 40 cm.

- i) In radians find the angle through which the pendulum swings.
- ii) Find the shortest distance between the maximum positions of the pendulum.

1**2**

- c) The number of bacteria N a person has after being infected with a virus after t hours is given by:

$$N = 10000e^{0.05t}$$

- i) Find the number of bacteria after 10 hours
- ii) Find the time required for the number of bacteria to reach 100000
- iii) At what rate is the bacteria increasing after 1 day

1**2****1**

- d) The area bounded by $y^2 = 3 - 2x - x^2$, $y \geq 0$ and between $x = -3$ and $x = 1$ is revolved about the x axis. Calculate the volume of the solid formed if this area is rotated about the x axis.

2**End of Question 7**

Question 8 (12 Marks)

Use a Separate Booklet

Marks

- a) Tiarn borrows \$500 000 to buy a house. An interest rate of 9% p.a. compounded monthly is charged on the outstanding balance. The loan is to be repaid in equal monthly instalments (R) over a 25 year period.

- i) Show the amount owing after 3 months is:

2

$$A_3 = 500000 \cdot 1.0075^3 - R[1 + 1.0075 + 1.0075^2]$$

- ii) Assuming this pattern continues the monthly repayment can be calculated using:

2

$$A_n = 500000 \cdot 1.0075^n - R[1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1}]$$

How much should Tiarn be paying each month?

- iii) How much interest does Tiarn pay over the 25 years?

1

- iv) What is the equivalent simple interest rate of this loan?

1

- b) i) Sketch the Parabola, whose focus is the point (2,5) and whose directrix is the line $y = -3$. Indicate on your diagram the vertex and its coordinates

2

- ii) Find the equation of the parabola.

1

- c) If $f(x) = 4 - 2^{-x}$ find:

- i) $f(x^2)$

1

- ii) $f(x)^2$

1

- iii) Is $f(x)$ even, odd or neither

1**End of Question 8**

Question 9 (12 Marks)

Use a Separate Booklet

Marks

- a) The acceleration $a \text{ ms}^{-2}$ of a moving particle is given after t seconds by $a = -2$. Initially the particle is located at $x = -3$ and its velocity is 4 ms^{-1}
- i) Find the velocity(v) and displacement (x) as functions of time (t) 2
 - ii) Determine when the particle is at rest. 2
 - iii) When will the particle first be at the origin? 2
 - iv) Sketch displacement (x) as a function of time (t) 2
- b) i) Differentiate $y = 3^{4x-2}$ with respect to x 3
- ii) Hence find: $\int 3^{4x-2} dx$ 1

End of Question 9

Question 10 (12 Marks)

Use a Separate Booklet

Marks

- a) A swimming pool is to be emptied for maintenance. The quantity of water Q litres, remaining in the pool at anytime, t minutes, after it starts to empty is given by:

$$Q(t) = 2000(25 - t)^2, \quad t \geq 0$$

- | | | |
|------|---|----------|
| i) | At what rate is the pool being emptied at any time (t) | 1 |
| ii) | How long will it take to half empty the pool to the nearest minute? | 2 |
| iii) | At what time is the water flowing out at 20 kL / minute. | 2 |
| iv) | What is the average water flow in the first 10 minutes in litres? | 2 |
- b) Adam is on a paddle board in the ocean 3 kilometres from the nearest point O on a straight beach. He needs to meet his friend Josh who is 6 kilometres along the beach from O. Adam is able to paddle at a rate of 4km/h and walk at a rate of 5km/h.

- | | | |
|-----|---|----------|
| i) | Draw a diagram to represent this information. | 1 |
| ii) | Show the total time $T(x)$ hours, for Adam to reach Josh is given by: | 2 |

$$T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$$

- | | | |
|------|--|----------|
| iii) | Find the minimum time for Adam to reach Josh on the beach. | 2 |
|------|--|----------|

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



International
Grammar School

Mathematics

Trial Higher School Certificate Examination

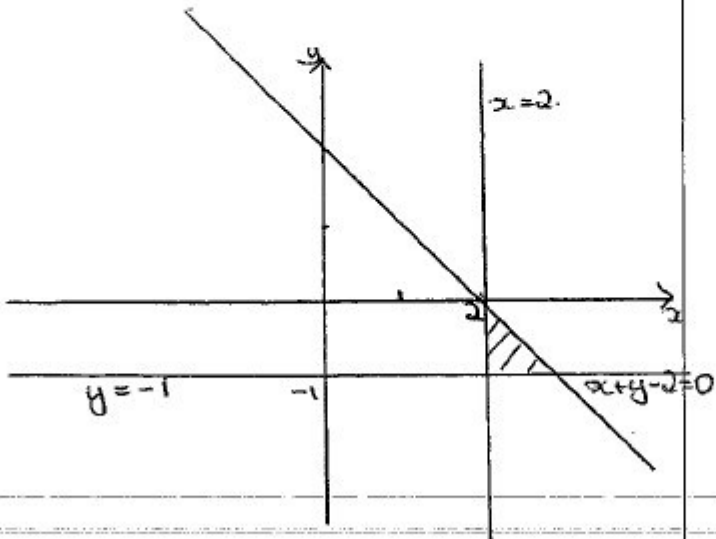
2010

SOLUTIONS

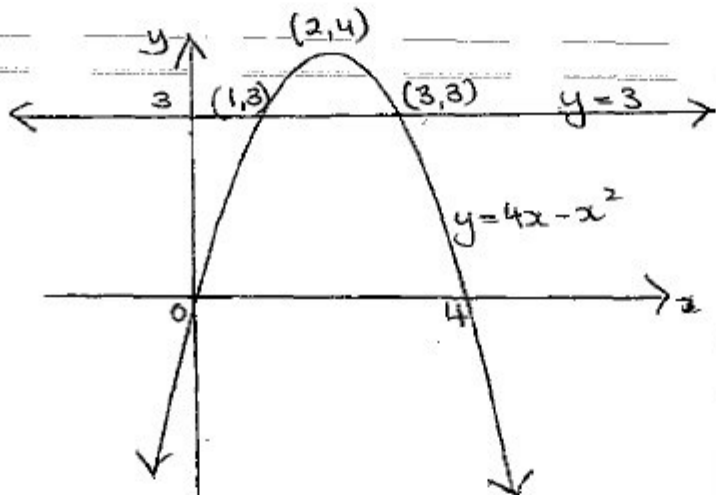
Question 1		Trial HSC Examination - Mathematics	2010
Part	Solution	Marks	Comment
a)	$\frac{\sqrt{5^2 + 144}}{13 - 6} = 4.9135...$ $= 4.91 \text{ 2dp}$	1	
b)	$\int_1^5 (3x - 7) dx$ $= \left[\frac{3x^2}{2} - 7x \right]_1^5$ $= \left(\frac{75}{2} - 35 \right) - \left(\frac{3}{2} - 7 \right)$ $= 8$	1	
c)	$\frac{1}{2 - \sqrt{3}} + \frac{1}{2 + \sqrt{3}}$ $= \frac{(2 + \sqrt{3}) + (2 - \sqrt{3})}{(2 - \sqrt{3})(2 + \sqrt{3})}$ $= \frac{2 + \sqrt{3} + 2 - \sqrt{3}}{4 - 3}$ $= 4$	2	
d)	i) $A = \frac{1}{2} ab \sin C$ $A = \frac{1}{2} \times 40 \times 80 \times \sin 110^\circ$ $= 1503.5 \text{ km}^2$ ii) $a^2 = b^2 + c^2 - 2bc \cos A$ $a^2 = 40^2 + 80^2 - 2 \times 40 \times 80 \times \cos 110^\circ$ $a^2 = 10188.93$ $a = 100.9$ $a = 101 \text{ km}$	4	1 correct use sine rule with correct angle 1 correct answer 1 correct use cosine rule 1 correct rounded answer

Question 1		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
e)	$\frac{5}{2a+6} + \frac{a}{a^2-9}$ $= \frac{5}{2(a+3)} + \frac{a}{(a+3)(a-3)}$ $= \frac{5(a-3) + 2a}{2(a+3)(a-3)}$ $= \frac{5a-15+2a}{2(a+3)(a-3)}$ $= \frac{7a-15}{2(a^2-9)}$	2	1	
f)	$ 2x+5 < 3$ $2x+5 < 3$ or $2x+5 > -3$ $2x < -2$ $2x > -8$ $x < -1$ $x > -4$ <u>check</u> $\therefore -4 < x < -1$	2	1	
		/12		

Question 2		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	i) $\frac{d}{dx}(3x^2 + 7)^6$ $= 6 \times 6x(3x^2 + 7)^5$ $= 36x(3x^2 + 7)^5$	2	1	1
	ii) $4x^2 e^{3x^3}$ $u = 4x^2 \quad v = e^{3x^3}$ $u' = 8x \quad v' = 9x^2 e^{3x^3}$ $\frac{d}{dx} = 8x e^{3x^3} + 4x^2 \times 9x^2 e^{3x^3}$ $= 8x e^{3x^3} + 36x^4 e^{3x^3}$ $= 4x e^{3x^3} [2 + 9x^3]$	2	1	1
	iii) $\frac{\pi \cos x}{x^2}$ $u = \pi \cos x \quad v = x^2$ $u' = -\pi \sin x \quad v' = 2x$ $\frac{d}{dx} = \frac{-\pi x^2 \sin x - 2\pi x \cos x}{(x^2)^2}$ $= \frac{\pi x(-x \sin x - 2 \cos x)}{x^4}$ $= \frac{\pi(-x \sin x - 2 \cos x)}{x^3}$	2	1	1
b)	$\int \frac{dx}{3x+5}$ $= \frac{1}{3} \int \frac{3}{3x+5} dx$ $= \frac{1}{3} \ln(3x+5) + C$	2	1	1

Question 2		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
c)	$y+1 \geq 0$ $x+y-2 \leq 0$ $x \geq 2$ $y \geq -1$ $y \leq 2-x$ 	2	1 correct lines 1 correct region.	
d)	$m = \tan \theta$ $-2.5 = \tan \theta$ $\theta = -68^{\circ}12'$ $\therefore \theta = 180^{\circ} - 68^{\circ}12'$ $= 111^{\circ}48'$	2	1 1	
		/12		

Question 3		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	$\lim_{x \rightarrow -3} \frac{x^2 + 8x + 15}{x + 3}$ $= \lim_{x \rightarrow -3} \frac{(x+3)(x+5)}{(x+3)}$ $= \lim_{x \rightarrow -3} x + 5$ $= 2.$	2	1	
b)	$\int_0^{\frac{\pi}{6}} x^2 + \sin 2x \, dx$ $= \left[\frac{x^3}{3} - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}}$ $= \left[\frac{\left(\frac{\pi}{6}\right)^3}{3} - \frac{1}{2} \cos\left(\frac{2\pi}{6}\right) \right] - \left[-\frac{1}{2} \cos 0 \right]$ $= \frac{\pi^3}{648} - \frac{1}{4} + \frac{1}{2}$ $= 0.298$	2	1	

Question 3	Trial HSC Examination - Mathematics	2010
Part	Solution	Marks Comment
c)	i) $y = 4x - x^2$ $y = 3$ Pts of intersection: $4x - x^2 = 3$ $x^2 - 4x + 3 = 0$ $(x-1)(x-3) = 0$ $x = 1 \quad x = 3$ at $x = 1, y = 4 \cdot 1 - 1 = 3 \quad (1, 3)$ $x = 3, y = 12 - 9 = 3 \quad (3, 3)$ 	2 1 correct diagram 1 all intercepts.
	ii) $A = \int_1^3 (4x - x^2 - 3) \cdot dx$ $= \left[2x^2 - \frac{x^3}{3} - 3x \right]_1^3$ $= \left(2 \cdot 9 - \frac{27}{3} - 3 \cdot 3 \right) - \left(2 \cdot 1 - \frac{1}{3} - 3 \right)$ $= (18 - 9 - 9) - \left(-1 - \frac{1}{3} \right)$ $= \frac{4}{3} \text{ u}^2$	2 1 correct integral 1 correct answer

Question 3		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
d)	$x + y + 3 = 0 \Rightarrow y = -x - 3$ $y = 2x^2 + 3x - 1$ If tangent, one point intersection. $2x^2 + 3x - 1 = -x - 3$ $2x^2 + 4x + 2 = 0$ $x^2 + 2x + 1 = 0$ $(x + 1)^2 = 0$ $x = -1$ Since there is one point of intersection, $x + y + 3 = 0$ is a tangent to $2x^2 + 3x - 1 = 0$ at $x = -1$.	2.	1	
e)	Amplitude = 1 Period = $\frac{2\pi}{\pi} = 2$	2	1	1
		/12		

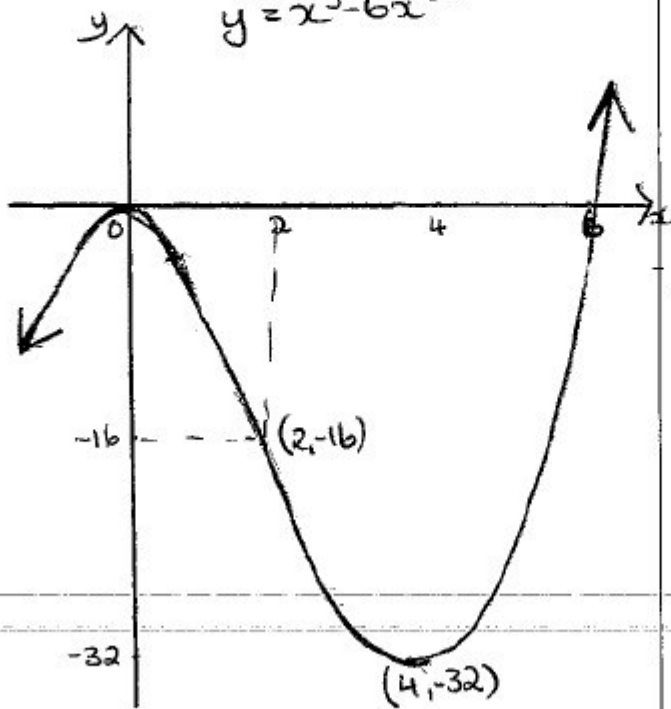
Question 4	Trial HSC Examination - Mathematics	2010
Part	Solution	Marks
a)	i) $AB = \sqrt{(4-0)^2 + (0+2)^2}$ $= \sqrt{16+4}$ $= \sqrt{20} = 2\sqrt{5} \text{ u.}$ $m_{AB} = \frac{0-2}{4-0}$ $= -\frac{2}{4} = -\frac{1}{2}$ ii) L parallel to AB so $m_L = \frac{1}{2}$. $y - 4 = \frac{1}{2}(x - 6)$ $2y + 8 = x - 6$ $x - 2y - 14 = 0$ iii) For D, $y = 0$. $x - 14 = 0$ $x = 14$ $D(14, 0)$ iv) $B(4, 0)$ $L: x - 2y - 14 = 0$ $d = \frac{ 4 - 2(0) - 14 }{\sqrt{1 + (-2)^2}} = \frac{ 4 - 14 }{\sqrt{5}}$ $= \frac{10}{\sqrt{5}} = 2\sqrt{5} \text{ u.}$ v) Area $ABDC$ (trapezium) $= \frac{h}{2}(a+b)$. Need $CD = \sqrt{(6-14)^2 + (-4-0)^2}$ $= \sqrt{8^2 + 4^2}$ $= \sqrt{80} = 4\sqrt{5}$ $A = \frac{2\sqrt{5}}{2}(2\sqrt{5} + 4\sqrt{5}) = \sqrt{5} \cdot 6\sqrt{5} = 30 \text{ u}^2$	2
b)	i) 2, 7, 12, 17, ... $T_n = 5n - 3$ ii) $T_{23} = 5 \times 23 - 3$ $= 112$ iii) $S_n = \frac{n}{2}[2a + (n-1)d]$ $= \frac{47}{2}[2 \times 2 + (47-1)5]$ $= 5499$	3

Question 4		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
c)	$\sec x + 1 = 3 \qquad 0 \leq x \leq \frac{\pi}{2}$ $\sec x = 2 \quad Q1$ i.e. $\frac{1}{\cos x} = 2.$ $\cos x = \frac{1}{2}$ $x = \pi/3 \quad 0 \leq x \leq \frac{\pi}{2}$	2	 1 1	

Question 5		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	$\int_0^{\frac{\pi}{4}} \cos 2x$ $= \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$ $= \frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0$ $= \frac{1}{2} \times 1$ $= \frac{1}{2} \text{ unit}^2$	2	1	1
b)	$2x^2 - 7x + 12 = 0$ $\alpha + \beta = \frac{-b}{a} \quad \alpha\beta = \frac{c}{a}$			
i)	$\alpha + \beta = \frac{-b}{a} = \frac{- -7}{2}$ $= \frac{7}{2}$	1		
ii)	$\alpha\beta = \frac{c}{a} = \frac{12}{2}$ $= 6$	1		
iii)	$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $= \frac{\left(\frac{7}{2}\right)}{6}$ $= \frac{7}{12}$	2	1	1
iv)	$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= \left(\frac{7}{2}\right)^2 - 2 \times 6$ $= \frac{1}{4}$	2	1	1

Question 5		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
c)	$a = 40 \quad r = \frac{5}{6} \quad r < 1$ G.P. $S_{\infty} = \frac{a}{1-r}$ $= \frac{40}{1 - \frac{5}{6}}$ $= \frac{40}{\frac{1}{6}}$ $= 240 \text{ cms}$	2	1 1	
d)	$y' = \frac{4x}{x^2+1}$ $y = 2 \int \frac{2x}{x^2+1} dx$ $= 2 \ln[x^2+1] + c$ when $x=0 \quad y=e$ $e = 2 \ln[0+1] + c$ $\therefore c = e$ $\therefore y = 2 \ln[x^2+1] + e$	2	1 1	
		/12		

Question 6	Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment
a)	Let $\angle ADB = x^\circ$ $\triangle ADB$ is isosceles ($AB = AD$) $\angle ADB = \angle ABD$ (base angles of $\triangle ADB$) then $2x + 100 = 180^\circ$ (angle sum $\triangle$) so $x = 40^\circ$ $\therefore \angle ADB = 40^\circ$ then $\angle DBC = 40^\circ$ (alt $\angle$'s = $AD \parallel BC$) $\therefore \triangle DBC$ is isosceles ($DB = DC$) $\therefore \angle DBC = 40^\circ$ (base $\angle$'s $\triangle DBC$) $\therefore \angle BDC = 180^\circ - 40^\circ - 40^\circ$ (angle sum $\triangle DBC$) $= 100^\circ$ $\therefore \angle BDC = 100^\circ$	3	1
b)	$f(x) = x^3 - 6x^2$ i) $x^3 - 6x^2 = 0$ $x^2(x - 6) = 0$ $x = 0, 6$ $\therefore$ crosses axes $(0, 0)$ $(6, 0)$ ii) $f'(x) = 3x^2 - 12x$ SP where $f'(x) = 0$ $3x(x - 4) = 0$ $x = 0, 4$ $y = 0, -32$ Nature: $f''(x) = 6x - 12$ at $(0, 4)$ $f''(x) < 0 \therefore$ max $(0, 4)$ $(4, -32)$ $f''(x) > 0 \therefore$ min. $(4, -32)$ so max TP at $(0, 4)$ min TP at $(4, -32)$ iii) POI when $f''(x) = 0$ and concavity changes. $f''(x) = 0 \quad 6x - 12 = 0$ $x = 2$ $y = -16$ at $x = 1, f''(x) < 0$ $x = 3, f''(x) > 0$ $\therefore$ concavity changes. $\therefore$ POI at $(2, -16)$	2	2.
		2	must use a test for nature
		1	2 (1 each)
			1 mark but must test for concavity.

Question 6		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
iv)	$y = x^3 - 6x^2$ 	3	1 shape 2 all pts marked (intercepts, TP and POI) [1 mark if intercepts and TP]	
v)	$y = f(x)$ increasing for $x < 0, x > 4$.	1	1	
		/12		

Question 7		Trial HSC Examination - Mathematics					2010													
Part	Solution						Marks	Comment												
a)	<table><tr><td>x</td><td>0</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>$f(x)$</td><td>2</td><td>1.936</td><td>1.732</td><td>1.323</td><td>0</td></tr></table>	x	0	0.5	1	1.5	2	$f(x)$	2	1.936	1.732	1.323	0						1	1
	x	0	0.5	1	1.5	2														
	$f(x)$	2	1.936	1.732	1.323	0														
$\int_a^b f(x)dx \approx \frac{h}{3} [(y_0 + y_n) + (4 \times \text{odds}) + (2 \times \text{evens})]$ $\approx \frac{0.5}{3} [(2 + 0) + 4(1.936 + 1.323) + 2(1.732)]$ $\approx 3.083 \text{ or } 3\frac{1}{12}$						2	1													
							1													
b)	i) $l = r\theta$ $40 = 50\theta$ $\theta = \frac{40}{50}$ $\theta = 0.8 \text{ radians}$						1	1												
	ii) $a^2 = b^2 + c^2 - 2bc \cos A$ $a^2 = 50^2 + 50^2 - 2 \times 50 \times 50 \times \cos \frac{4}{5}$ $a = 38.94 \text{ cm}$						2	1												
							1													
c)	i) $N = 10000e^{0.05t}$ $N = 10000e^{0.05 \times 10}$ $N = 16487$						1	1												
	ii) $100000 = 10000e^{0.05t}$ $10 = e^{0.05t}$ $\ln 10 = \ln(e^{0.05t})$ $\ln 10 = 0.05t(\ln e)$ $t = \frac{\ln 10}{0.05}$ $= 46 \text{ hours}$						2	1												
	iii) $N' = 500e^{0.05t}$ <i>when $t = 24 \text{ hours}$</i> $N' = 500e^{0.05 \times 24}$ $= 1660 \text{ bacteria/hour}$						1	1												

Question 7		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
d)	$y^2 = 3 - 2x - x^2$ $V = \pi \int_a^b y^2 dx$ $V = \pi \int_{-3}^1 3 - 2x - x^2 dx$ $= \pi \left[3x - x^2 - \frac{x^3}{3} \right]_{-3}^1$ $= \pi \left[\left(3 - 1 - \frac{1}{3} \right) - (-9 - 9 + 9) \right]$ $= \frac{32\pi}{3}$	2	1	1
		/12		

Question 8		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	i) $r = 9 \div 100 \div 12$ $r = 0.0075$ $A = P(1 + r)^n$ $A_1 = 500000(1.0075)^1 - R$ $A_2 = A_1(1.0075)^1 - R$ $A_2 = 500000(1.0075)^2 - R(1.0075) - R$ $A_3 = A_2(1.0075)^1 - R$ $A_3 = 500000[(1.0075)^2 - R(1.0075) - R](1.0075) - R$ $= 500000(1.0075)^3 - R(1.0075)^2 - R(1.0075) - R$ $= 500000(1.0075)^3 - R[1 + 1.0075 + 1.0075^2]$ as required	2	1	1
	ii) $A_n = 0 \text{ as all money is repaid}$ $\therefore 0 = 500000(1.0075)^{300} -$ $R[1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1}]$ $R = \frac{500000(1.0075)^{300}}{[1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1}]}$ geometric series with $a = 1, r = 1.0075, n = 300$ $S_n = \frac{a(r^n - 1)}{r - 1}$ $S_{300} = \frac{1(1.0075^{300} - 1)}{1.0075 - 1}$ $S_{300} \approx 1121.121937$ $R = \frac{500000(1.0075)^{300}}{S_{300}}$ $R = \$4195.98$	2	1	1
	iii) Total repaid = $\\$4195.98 \times 300$ $= \$1258794.00$ Interest = $\\$1258794 - 500000$ $= \$758794.00$	1		1

Question 8		Trial HSC Examination - Mathematics	2010
Part	Solution	Marks	Comment
a)	iv) $SI = Prn$ $758794 = 500000 \times r \times 25$ $r = 6.07\%$	1	1
b)	i) A hand-drawn graph of a parabola opening upwards on a Cartesian coordinate system. The x-axis has tick marks from 1 to 5. The y-axis has tick marks from -3 to 6. The vertex of the parabola is labeled as (2, 1), which lies on the x-axis. Another point on the curve is labeled as (2, 5). Below the x-axis, there is a horizontal line segment labeled y = -3. ii) $(x-2)^2 = 4 \cdot 4(y-1)$ $= 16(y-1)$	2. 1 1	1 correct shape 2 correct vertex
c)	i) $f(x) = 4 - 2^{-x}$ $f(x^2) = 4 - 2^{-x^2}$ ii) $[f(x)]^2 = [4 - 2^{-x}] \times [4 - 2^{-x}]$ $= 16 - 2^3 \times 2^{-x} + (2^{-x})^2$ $= 16 - 2^{3-x} + 2^{-2x}$ iii) $f(-x) = 4 - 2^{-(-x)}$ $= 4 - 2^x$ $\neq f(x) \text{ or } -f(x)$ $\therefore$ the function is neither odd nor even	1 1 1 1	1 1 1 1
		/12	

Question 9		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	i) $a = -2 \quad x = -3 \quad v = 4ms^{-1}$ $a = -2$ $v = \int -2dt$ $= -2t + c$ <i>when $t = 0 \quad v = 4$</i> $4 = -2 \times 0 + c$ $c = 4$ $\therefore v = -2t + 4$ $x = \int -2t + 4dt$ $= -t^2 + 4t + c$ <i>when $t = 0 \quad x = -3$</i> $-3 = 0 + 0 + c$ $c = -3$ $\therefore x = -t^2 + 4t - 3$	2	1	
	ii) Particle at rest when $v=0$ $v = -2t + 4$ $0 = -2t + 4$ $2t = 4$ $t = 2 \text{ sec onds}$ $\therefore \text{particle at rest when } t = 2 \text{ sec onds}$	2	1	
	iii) Particle at the origin when $x = 0$ $x = -t^2 + 4t - 3$ $0 = -t^2 + 4t - 3$ $0 = -(t^2 - 4t + 3)$ $0 = -(t-3)(t-1)$ $\therefore t = 1 \text{ or } 3 \text{ sec onds}$ particle first at the origin when $t = 1 \text{ sec ond}$	2	1	

Question 9		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
	ii) $\int 3^{4x-2} dx$ $= \frac{1}{4 \ln 3} \int 4 \ln 3 (3^{4x-2})$ $= \frac{1}{4 \ln 3} \times (3^{4x-2}) + c$ $= \frac{(3^{4x-2})}{4 \ln 3} + c$	1	1	
		/12		

Question 10		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
a)	i) $Q(t) = 2000(25 - t)^2, t \geq 0$ $Q'(t) = -4000(25 - t)$ $\therefore$ it is emptying at a rate of $4000(25 - t)$ litres/minute	1	1	
	ii) Pool full at $t = 0$ $Q(t) = 2000(25 - 0)^2$ $= 1250000$ litres $\therefore$ half full = 625000 litres $625000 = 2000(25 - t)^2$ $312.5 = 625 - 50t + t^2$ $t^2 - 50t + 312.5 = 0$	2	1	
	$2t^2 - 100t + 625 = 0$ $t = \frac{-(-100) \pm \sqrt{100^2 - 4 \times 2 \times 625}}{2 \times 2}$ $t = \frac{100 \pm \sqrt{5000}}{4}$ $t = \frac{100 \pm 50\sqrt{2}}{4}$ $t = \frac{2(50 \pm 25\sqrt{2})}{4}$ $t = \frac{50 \pm 25\sqrt{2}}{2}$ $t = 7.322$ or 42.68 $\therefore t = 7$ minutes $\therefore$ it will take ≈ 7 minutes to half empty the pool			
	iii) $20kL = 20000L / \text{min}$ $20000 = -4000(25 - t)$ $20000 = -100000 + 4000t$ $4000t = 120000$ $t = 30 \text{ min}$ $\therefore$ the flow rate will be 20kL after 30 minutes	2	1	1

Question 10		Trial HSC Examination - Mathematics		2010
Part	Solution	Marks	Comment	
b)	iii) $T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$ $T'(x) = \frac{x}{4\sqrt{x^2 + 9}} - \frac{1}{5}$ $= \frac{5x - 4\sqrt{x^2 + 9}}{20\sqrt{x^2 + 9}}$ Min when $T'(x) = 0$ $0 = \frac{5x - 4\sqrt{x^2 + 9}}{20\sqrt{x^2 + 9}}$ $0 = 5x - 4\sqrt{x^2 + 9}$ $5x = 4\sqrt{x^2 + 9} \text{ (square both sides)}$ $25x^2 = 16x^2 + 144$ $9x^2 = 144$ $x^2 = 16$ $x = \pm 4 \text{ (} x \neq -4 \text{)}$ $\therefore x = 4$ check minimum when $x < 4, T'(x) < 0$ when $x > 4, T'(x) > 0$ $\therefore$ minimum at $x = 4$ $\therefore$ Adam paddles to C - 4 kilometres from O $T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$ $T(4) = \frac{\sqrt{4^2 + 9}}{4} + \frac{6 - 4}{5}$ $= 1.65 \text{ hours}$ $= 1 \text{ hour \& } 39 \text{ min s}$	2	1	
		/12		